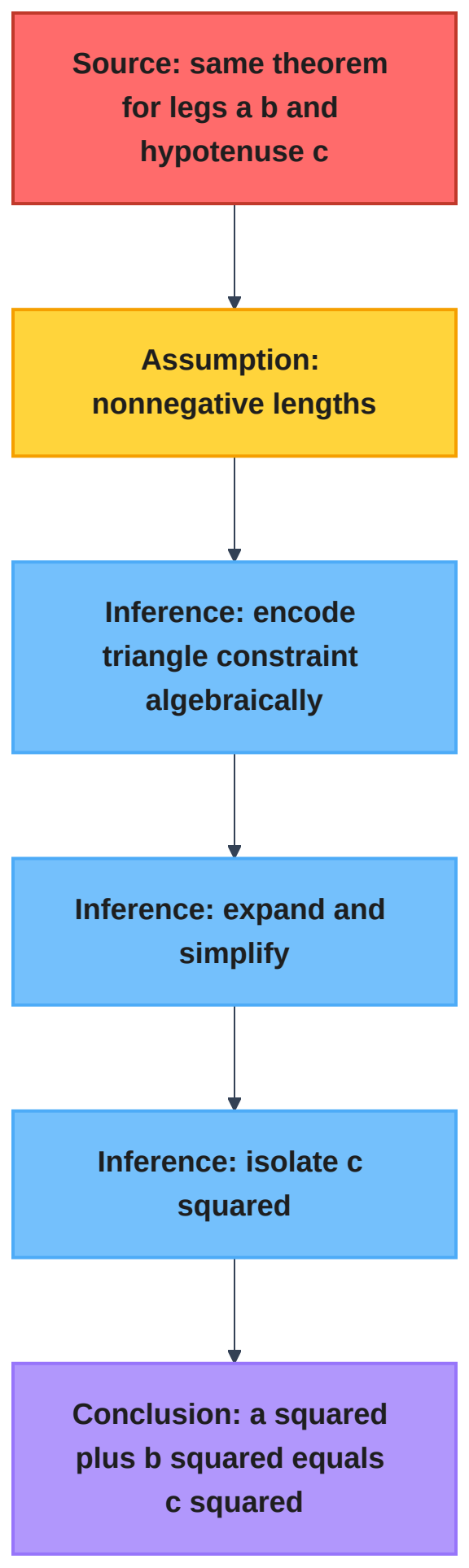


**Source: same theorem
for legs a b and
hypotenuse c**



```
graph TD; A[Source: same theorem for legs a b and hypotenuse c] --> B[Assumption: nonnegative lengths]; B --> C[Inference: encode triangle constraint algebraically]; C --> D[Inference: expand and simplify]; D --> E[Inference: isolate c squared]; E --> F[Conclusion: a squared plus b squared equals c squared];
```

**Assumption:
nonnegative lengths**

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triangle constraint
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**Conclusion: a squared
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